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M-type hexaferrite comprising antimony

Abstract

In an aspect, a ferrite composition can comprise a SbCo—M-type ferrite having the formula: $\text{Me}_{1-x}\text{Sb}_x\text{Co}_y\text{M}'_{1-y}\text{Fe}_{12-x-2y}\text{O}_{19}$, wherein Me is at least one of Sr, Pb, or Ba; M' is at least one of Ti, Zr, Ru, or Ir; x is 0.001 to 0.3; and y is 0.8 to 1.3. In another aspect, a method of making the ferrite composition comprises mixing ferrite precursor compounds comprising Me, Fe, Sb, Co, and M'; and sintering the ferrite precursor compounds in an oxygen atmosphere to form the SbCo—M-type ferrite. In yet another aspect, a composite comprises the ferrite composition and a polymer. In still another aspect, an article comprises the ferrite composition.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a National Stage application of PCT/US2020/057827, filed Oct. 29, 2020, which claims the benefit of U.S. Provisional Application No. 62/927,783, filed Oct. 30, 2019, both of which are incorporated by reference in their entirety herein.

BACKGROUND

(1) Improved performance and miniaturization are needed to meet the ever-increasing demands of devices used in very high frequency (VHF) applications, which are of particular interest in a variety of commercial and defense related industries. As an important component in radar and modern wireless communication systems, antenna elements with compact sizes are constantly being developed. It has been challenging however to develop ferrite materials for use in such high frequency applications as most ferrite materials exhibit relatively high magnetic loss at high frequencies. In general, hexagonal ferrites, or hexaferrites, are a type of iron-oxide ceramic compound that has a hexagonal crystal structure and exhibits magnetic properties. Several types of families of hexaferrites are known, including Z-type ferrites, Ba₃Me₂Fe₂₄O₄₁, and Y-type ferrites, Ba₂Me₂Fe₁₂O₂₂, where Me can be a small 2+ cation such as Co, Ni, or Zn, and Sr can be substituted for Ba. Other hexaferrite types include M-type ferrites ((Ba,Sr)Fe₁₂O₁₉), W-type ferrites ((Ba,Sr)Me₂Fe₁₆O₂₇), X-type ferrites ((Ba,Sr)₂Me₂Fe₂₈O₄₆), and U-type ferrites ((Ba,Sr)₄Me₂Fe₃₆O₆₀).

(2) Despite improvements in ferrite composites, their use in high frequency and microwave devices

is limited as these substitutions also result in an increase in the magnetic loss. Improved ferrite compositions are therefore desired.

BRIEF SUMMARY

(3) Disclosed herein is a SbCo—M-type phase ferrite.

(4) In an aspect, a ferrite composition comprises a SbCo—M-type ferrite having the formula: $\text{Me}_{1-x}\text{Sb}_x\text{Co}_y\text{M}'_{1-y}\text{Fe}_{12-2x}\text{O}_{19}$, wherein Me is at least one of Sr, Pb, or Ba; M' is at least one of Ti, Zr, Ru, or Ir; x is 0.001 to 0.3; and y is 0.8 to 1.3.

(5) In another aspect, a method of making the ferrite composition comprises mixing ferrite precursor compounds comprising Me, Fe, Sb, Co, and M'; and sintering the ferrite precursor compounds in an oxygen atmosphere to form the SbCo—M-type ferrite.

(6) In yet another aspect, a composite comprises the ferrite composition and a polymer.

(7) In still another aspect, an article comprises the ferrite composition.

(8) The above described and other features are exemplified by the following FIGURE, detailed description, and claims.

Description

BRIEF DESCRIPTION OF THE DRAWING

(1) The following FIGURE is an exemplary embodiment that is provided to illustrate the present disclosure. The FIGURE is illustrative of the example, which is not intended to limit devices made in accordance with the disclosure to the materials, conditions, or process parameters set forth herein.

(2) FIG. 1 is a graphical illustration of the permeability and magnetic loss tangent with frequency for Example 1 and Example 2.

DETAILED DESCRIPTION

(3) It was discovered that substituting at least a portion of the metal ion in an M-type hexaferrite with antimony could result in a SbCo—M-type ferrite with improved properties. Specifically, the SbCo—M-type ferrite that can have the general formula:

$\text{Me}_{1-x}\text{Sb}_x\text{Co}_y\text{M}'_{1-y}\text{Fe}_{12-2x}\text{O}_{19}$

wherein Me is at least one of Sr, Pb, or Ba; M' is at least one of Ti, Zr, Ru, or Ir; x is 0.001 to 0.3; and y is 0.8 to 1.3. In the formula, x can be 0.005 to 0.1. In the formula, y can be 1 to 1.2. In the formula, Me can include Ba. In the formula, M' can include Ti.

(4) The SbCo—M-type ferrite that can have the formula:

$\text{Ba}_{1-x}\text{Sb}_x\text{Co}_y\text{Ti}'_{1-y}\text{Fe}_{12-2x}\text{O}_{19}$

wherein a portion, but not all of the Ba can be substituted with at least one of Sr or Pb; wherein a portion, but not all of the Ti can be substituted with at least one of Zr, Ru, or Ir; x is 0.001 to 0.3; and y is 0.8 to 1.3. In the formula, x can be 0.005 to 0.1. In the formula, y can be 1 to 1.2.

(5) The SbCo—M-type ferrite that can have the formula:

$\text{Ba}_{1-x}\text{Sb}_x\text{Co}_y\text{Ti}'_{1-y}\text{Fe}_{12-2x}\text{O}_{19}$

wherein x is 0.001 to 0.3, or 0.005 to 0.1; and y is 0.8 to 1.3, or 1 to 1.2.

(6) The SbCo—M-type ferrite can have a planar easy magnetization (c-plane). The SbCo—M-type ferrite can have at least one of a high permeability, high operating frequency, or a low magnetic loss, making it suitable for use as an antenna or an inductor at VHF (very high frequencies), for example, of 1 to 300 megahertz, or 30 to 300 megahertz. The SbCo—M-type ferrite can have a permeability of greater than or equal to 30, or greater than or equal to 50, or 28 to 55, or 28 to 35 at a frequency of 1 to 300 megahertz in the form of a solid ceramic. The SbCo—M-type ferrite can have a magnetic loss tangent tan δ of less than or equal to 0.3, less than or equal to 0.1, or 0.01 to 0.3 at a frequency of 1 to 300 megahertz in the form of a solid ceramic. The SbCo—M-type ferrite can have a resonant frequency of greater than or equal to 250, or 250 to 500 megahertz in the form

of a solid ceramic. The SbCo—M-type ferrite can have a Snoek product (the permeability measured at a frequency of 0.01 gigahertz times the resonant frequency) of greater than or equal to 8 gigahertz, or greater than or equal to 9 gigahertz, or 9 to 15 gigahertz in the form of a solid ceramic. The SbCo—M-type ferrite can have a complex permeability at 200 megahertz of 30 to 40, or 30 to 35. The SbCo—M-type ferrite can have a complex permeability at 300 megahertz of 20 to 35, or 35 to 30.

(7) As used herein, the magnetic complex permeability is measured using an E4991B Impedance Analyzer commercially available from Keysight Technologies (1 MHz to 1 GHz) with the 16454A test fixture. The applicable ferrite materials are toroidal cores with the dimensions: outer diameter of 16 millimeters, inner diameter of 9 millimeters, height of 3 millimeters.

(8) The SbCo—M-type ferrite can be prepared using any suitable method. Generally, the SbCo—M-type ferrite can be formed by first forming a mixture comprising the precursor compounds including oxides of at least Me, Sb, Co, M', and Fe. For example, the mixture can comprise at least MeCO.sub.3 , Sb.sub.2O.sub.3 , Co.sub.3O.sub.4 , M'O.sub.2 , and Fe.sub.2O.sub.3 . The mixture can comprise 5 to 20 weight percent of MeCO.sub.3 based on the total weight of the precursor compounds. The mixture can comprise 0.01 to 20 weight percent, or 5 to 20 weight percent of Sb.sub.2O.sub.3 based on the total weight of the precursor compounds. The mixture can comprise 1 to 15 weight percent of Co.sub.3O.sub.4 based on the total weight of the precursor compounds. The mixture can comprise 0.1 to 10 weight percent of M'O.sub.2 based on the total weight of the precursor compounds. The mixture can comprise 50 to 80 weight percent of Fe.sub.2O.sub.3 based on the total weight of the precursor compounds. The mixture can further comprise an oxide of at least one of Zn, Mg, Ti, Zr, Sn, or Cu.

(9) The mixture can be calcined in air to form a calcined ferrite comprising an M-type ferrite. The calcining can occur at a calcination temperature of 800 to 1,300 degrees Celsius ($^{\circ}\text{C.}$). The calcining can occur for a calcination time of 0.5 to 20 hours, 1 to 10 hours, or 2 to 5 hours. It is noted that the calcination step is optional and the only heating step can be the sintering.

(10) The calcined ferrite can be crushed and optionally sieved to form particulates. The sieving can comprise sieving through a number 10 to 300 mesh screen. Coarse particles can be further ground to reduce the particle size. The particulates can have a median D50 particle size by volume of 0.5 to 50 micrometers, or 0.5 to 10 micrometers. The particle size can be determined using Horiba LA-910 laser light scattering PSD analyzer or as determined in accordance with ASTM D4464-15. The calcined ferrite can be mixed with a binder. The binder can comprise at least one of poly(vinyl alcohol), poly(vinyl butyral), methyl cellulose, poly(ethylene glycol), or poly(alkylene carbonate). The binder can be present in an amount of 8 to 12 weight percent based on the total amount of the calcined ferrite plus the binder. The binder can facilitate shaping of the calcined ferrite and can fully burn off during sintering. The bound calcined ferrite can have a median D50 particle size by volume of 10 to 300 micrometers, or 50 to 300 micrometers.

(11) The calcined ferrite can be optionally shaped and sintered to form the SbCo—M-type ferrite. The method of shaping is not limited and can comprise at least one of uniaxial compression, isostatic pressing, casting, pressing, or the like. The pressing can occur at a pressure of 0.5 to 2 metric tons per centimeter squared. The sintering can occur at a sintering temperature of 1,000 to 1,300 $^{\circ}\text{C.}$, or 1,200 to 1,250 $^{\circ}\text{C.}$ The ramping temperature up to and down from the sintering temperature can each independently occur at a ramp rate of 1 to 5 $^{\circ}\text{C.}$ per minute. The sintering can occur for a sintering time of 1 to 20 hours, or 5 to 12 hours. The sintering can be performed in an oxygen environment to help reduce the dielectric loss. Oxygen can be introduced to the sintering chamber at a flow rate of 0.1 to 10 liters per minute. The sintered SbCo—M-type ferrite can be finished in any desired manner, for example, by cutting or polishing, depending on the application.

(12) Prior to the sintering, an amount of bismuth oxide can be added to the calcined ferrite. The bismuth oxide can be added in an amount of 0.01 to 5 weight percent based on the total weight of the calcined ferrite. Bi.sub.2O.sub.3 can be added into a starting mixture or to the calcined ferrite

powder, depending upon the desired distribution of Bi ions. Bi ions can be located in the crystalline lattice or can be at the grain boundaries. The addition of an additional amount of Bi.sub.2O.sub.3, which can be present at grain boundaries of the M-type ferrite phase, can aid in achieving a composition that has equal or substantially equal values of permeability and permittivity while retaining low magnetic and dielectric loss tangents.

(13) The SbCo—M-type ferrite can have an average grain size of 1 to 100 micrometers, or 5 to 50 micrometers. The average grain size can be measured using conventional methods, for example, by transmission electron microscopy or field emission scanning electron microscopy.

(14) The SbCo—M-type ferrite can be a bulk ceramic or can be present in a composite, for example, comprising the SbCo—M-type ferrite and a polymer. The polymer can comprise a thermoplastic or a thermoset. As used herein, the term “thermoplastic” refers to a material that is plastic or deformable, melts to a liquid when heated, and freezes to a brittle, glassy state when cooled sufficiently. Examples of thermoplastic polymers that can be used include cyclic olefin polymers (including polynorbornenes and copolymers containing norbornenyl units, for example, copolymers of a cyclic polymer such as norbornene and an acyclic olefin such as ethylene or propylene), fluoropolymers (for example, polyvinyl fluoride (PVF), polyvinylidene fluoride (PVDF), fluorinated ethylene-propylene (FEP), polytetrafluoroethylene (PTFE), poly(ethylene-tetrafluoroethylene (PETFE), or perfluoroalkoxy (PFA)), polyacetals (for example, polyoxyethylene and polyoxymethylene), poly(C.sub.1-6 alkyl)acrylates, polyacrylamides (including unsubstituted and mono-N— or di-N—(C.sub.1-8 alkyl)acrylamides), polyacrylonitriles, polyamides (for example, aliphatic polyamides, polyphthalamides, or polyaramides), polyamideimides, polyanhydrides, polyarylene ethers (for example, polyphenylene ethers), polyarylene ether ketones (for example, polyether ether ketones (PEEK) and polyether ketone ketones (PEKK)), polyarylene ketones, polyarylene sulfides (for example, polyphenylene sulfides (PPS)), polyarylene sulfones (for example, polyethersulfones (PES), polyphenylene sulfones (PPS), and the like), polybenzothiazoles, polybenzoxazoles, polybenzimidazoles, polycarbonates (including homopolycarbonates or polycarbonate copolymers such as polycarbonate-siloxanes, polycarbonate-esters, or polycarbonate-ester-siloxanes), polyesters (for example, polyethylene terephthalates, polybutylene terephthalates, polyarylates, or polyester copolymers such as polyester-ethers), polyetherimides (for example, copolymers such as polyetherimide-siloxane copolymers), polyimides (for example, copolymers such as polyimide-siloxane copolymers), poly(C.sub.1-6 alkyl)methacrylates, polyalkylacrylamides (for example, unsubstituted and mono-N— or di-N—(C.sub.1-8 alkyl)acrylamides), polyolefins (for example, polyethylenes, such as high density polyethylene (HDPE), low density polyethylene (LDPE), and linear low density polyethylene (LLDPE), polypropylenes, and their halogenated derivatives (such as polytetrafluoroethylenes), and their copolymers, for example, ethylene-alpha-olefin copolymers), polyoxadiazoles, polyoxymethylenes, polyphthalides, polysilazanes, polysiloxanes (silicones), polystyrenes (for example, copolymers such as acrylonitrile-butadiene-styrene (ABS) or methyl methacrylate-butadiene-styrene (MBS)), polysulfides, polysulfonamides, polysulfonates, polysulfones, polythioesters, polytriazines, polyureas, polyurethanes, vinyl polymers (for example, polyvinyl alcohols, polyvinyl esters, polyvinyl ethers, polyvinyl halides (for example, polyvinyl chloride), polyvinyl ketones, polyvinyl nitriles, or polyvinyl thioethers), a paraffin wax, or the like. A combination comprising at least one of the foregoing thermoplastic polymers can be used.

(15) Thermoset polymers are derived from thermosetting monomers or prepolymers (resins) that can irreversibly harden and become insoluble with polymerization or cure, which can be induced by heat or exposure to radiation (e.g., ultraviolet light, visible light, infrared light, or electron beam (e-beam) radiation). Thermoset polymers include alkyds, bismaleimide polymers, bismaleimide triazine polymers, cyanate ester polymers, benzocyclobutene polymers, benzoxazine polymers, diallyl phthalate polymers, epoxies, hydroxymethylfuran polymers, melamine-formaldehyde polymers, phenolics (including phenol-formaldehyde polymers such as novolacs and resoles),

benzoxazines, polydienes such as polybutadienes (including homopolymers and copolymers thereof, e.g. poly(butadiene-isoprene)), polyisocyanates, polyureas, polyurethanes, triallyl cyanurate polymers, triallyl isocyanurate polymers, certain silicones, and polymerizable prepolymers (e.g., prepolymers having ethylenic unsaturation, such as unsaturated polyesters or polyimides), or the like. The prepolymers can be polymerized, copolymerized, or crosslinked, e.g., with a reactive monomer such as styrene, alpha-methylstyrene, vinyltoluene, chlorostyrene, acrylic acid, (meth)acrylic acid, a (C.sub.1-6 alkyl)acrylate, a (C.sub.1-6 alkyl) methacrylate, acrylonitrile, vinyl acetate, allyl acetate, triallyl cyanurate, triallyl isocyanurate, or acrylamide.

(16) The SbCo—M-type ferrite can be mixed with a polymer to form the composite. The mixing can comprise mixing at a temperature of a melting point of the polymer. The mixing can comprise mixing a polymer precursor with the BiRu—Co.sub.2M ferrite and reacting the polymer precursor to form the polymer. Methods of forming the composite are not limited and can comprise injection molding, reaction injection molding, laminating, extruding, compression molding, calendaring, casting, and the like. The composite can be free of a void space.

(17) The SbCo—M-type ferrite can be surface-treated (also referred to herein as coated) to aid dispersion into the polymer or polymer precursor, for example, with at least one of a surfactant (such as oleylamine oleic acid), an inorganic material (such as SiO.sub.2, Al.sub.2O.sub.3, and MgO), a silane, a titanate, or a zirconate.

(18) The coating can comprise at least one of a silane coating, a titanate coating, or a zirconate coating. The coating can comprise at least one of phenyltrimethoxysilane, p-chloromethylphenyltrimethoxy silane, aminoethylaminotrimethoxy silane, aminoethylaminopropyltrimethoxy silane, phenyltriethoxysilane, 3,3,3-trifluoropropyl trimethoxysilane, (tridecafluoro-1,1,2,2-tetrahydrodecyl)-1-triethoxysilane, neopentyl(diallyl)oxytrineodecanoyl titanate, neopentyl(diallyl)oxytri(dioctyl)phosphate titanate, neopentyl(diallyl)oxytri(dioctyl)pyrophosphate zirconate, or neopentyl(diallyl) oxytri(N-ethylenediamino)ethyl zirconate. The coating can comprise a silane coating comprising at least one of an aromatic silane such as phenyltrimethoxysilane or a fluorinated aliphatic alkoxy silane such as (tridecafluoro-1,1,2,2-tetrahydrooctyl)triethoxy silane.

(19) An article can comprise the SbCo—M-type ferrite. The article can be an antenna or an inductor core. The article can be for use in the frequency 0.1 to 4 gigahertz range, or in the 0.5 to 2 gigahertz range. The article can be used for a variety of devices operable within the ultrahigh frequency range, such as a high frequency or microwave antenna, filter, inductor, circulator, or phase shifter. The article can be operable at frequencies greater than or equal to 1 GHz, or at frequencies 1 to 3 GHz. Such articles can be used in commercial and military applications, weather radar, scientific communications, wireless communications, autonomous vehicles, aircraft communications, space communications, satellite communications, or surveillance. The article can be used for EMI (electromagnetic interference) shielding applications.

(20) As disclosed herein, the SbCo—M-type ferrite can have the general formula: Me.sub.1-xSb.sub.xCo.sub.y+xM'.sub.yFe.sub.12-x-2yO.sub.19, wherein Me is at least one of Sr, Pb, or Ba; M' is at least one of Ti, Zr, Ru, or Ir; x is 0.001 to 0.3; and y is 0.8 to 1.3. In the formula, x can be 0.005 to 0.1. In the formula, y can be 1 to 1.2. In the formula, Me can include or can be Ba. In the formula, M' can include or can be Ti. The SbCo—M-type ferrite can have an average grain size of 1 to 100 micrometers, or 5 to 50 micrometers. The SbCo—M-type ferrite can have desirable permeability values over a high frequency range. The SbCo—M-type ferrite can be in the form of a bulk ceramic. A composite can comprise the SbCo—M-type ferrite and a polymer. The composite can be free of a void space. An article can comprise the SbCo—M-type ferrite. An article can comprise the composite. The article can be, for example, a microwave antenna, a filter, an inductor, a circulator, or a phase shifter. The article can be used in various applications including commercial and military applications, weather radar, scientific communications, wireless communications, autonomous vehicles, aircraft communications, space communications, satellite communications,

surveillance, or EMI shielding applications.

(21) The following examples are provided to illustrate the present disclosure. The examples are merely illustrative and are not intended to limit devices made in accordance with the disclosure to the materials, conditions, or process parameters set forth therein.

EXAMPLES

(22) In the examples, the magnetic permeability of the resulting ferrite samples was measured using an E4991B Impedance Analyzer commercially available from Keysight Technologies over a frequency of 1 MHz to 1 GHz.

(23) The particle sizes were determined using Horiba LA-910 laser light scattering PSD analyzer or as determined in accordance with ASTM D4464-15. The reported particle size is the median D50 particle size by volume.

Examples 1 and 2: Effect of Antimony on the Co—M-Type Phase Ferrite

(24) Two samples of the Co—M-type phase ferrite with and without antimony were formed by mixing appropriate amounts of BaCO₃, Sb₂O₃, Co₃O₄, TiO₂, and Fe₂O₃ in a wet planetary mill to form a mixture. The mixtures were calcined to a calcination temperature of 1,100° C. for a soak time of 4 hours in air. The calcined ferrite was crushed and sieved through a 40 mesh screen. The coarse particles were ground to form particulates having a median D50 particle size by volume of 0.5 to 10 micrometers in a wet planetary mill. A granulate mixture was formed by mixing the particulates with 0.5 to 5 weight percent of poly(vinyl alcohol) based the total amount of the granulate mixture. The granulate mixture was then compressed to form toroid granules having an outer diameter of 18 millimeters, an inner diameter of 10 millimeters, and a thickness of 3 to 3.5 millimeters at a pressure of 1 metric tons per centimeter squared. The toroid granules were sintered at a sintering temperature of 1,240° C. for 4 hours in air, using a ramp rate: 3° C. per minute and a cooling rate of −3° C. per minute. The Co—M-type phase ferrites with and without antimony had formulae

Ba_{0.1}Co_{1.1}Ti_{1.1}Fe_{9.8}O₁₉ and

Ba_{0.99}Sb_{0.01}Co_{1.1}Ti_{1.1}Fe_{9.79}O₁₉, respectively.

(25) The permeability (solid lines) and the magnetic loss tangent (dashed lines) are shown in FIG. 1 as a function of frequency and specific values of the different compositions are shown in Table 1, where fr stands for the resonant frequency.

(26) TABLE-US-00001 TABLE 1 Ferrite Sample 1 2 Antimony N Y μ' at 1 GHz 26 32 μ' at 100 GHz 27 30 tanδ.μ at 100 GHz 0.30 0.22 fr (MHz) 210 300 Snoek Product (GHz) 5.46 9.60

(27) The results show that incorporating an amount of antimony in the M-type ferrite, results in a significant increase in both the permeability and resonance frequency at high frequencies.

(28) Set forth below are non-limiting aspects of the present disclosure.

(29) Aspect 1: A ferrite composition comprising a SbCo—M-type ferrite having the formula Me_{1-x}Sb_xCo_{y+xM'}Fe_{12-x-2y}O₁₉, wherein Me is at least one of Sr, Pb, or Ba; M' is at least one of Ti, Zr, Ru, or Ir; x is 0.001 to 0.3; and y is 0.8 to 1.3.

(30) Aspect 2: The ferrite composition of Aspect 1, wherein y is 1 to 1.2.

(31) Aspect 3: The ferrite composition of any one or more of the preceding aspects, wherein Me is at least Ba.

(32) Aspect 4: The ferrite composition of any one or more of the preceding aspects, wherein M' is at least Ti.

(33) Aspect 5: The ferrite composition of any one or more of the preceding aspects, wherein x is 0.005 to 0.1.

(34) Aspect 6: The ferrite composition of any one or more of the preceding aspects, wherein the ferrite composition has a permeability of greater than or equal to 30, or greater than or equal to 50, or 28 to 55, or 28 to 35 at a frequency of 1 to 300 megahertz in the form of a solid ceramic.

(35) Aspect 7: The ferrite composition of any one or more of the preceding aspects, wherein the ferrite composition has a complex permeability at 200 megahertz of 30 to 40, or 30 to 35; or the

SbCo—M-type ferrite can have a complex permeability at 300 megahertz of 20 to 35, or 35 to 30.

(36) Aspect 8: The ferrite composition of any one or more of the preceding aspects, wherein the ferrite composition has a magnetic loss tangent $\tan \delta$ of less than or equal to 0.3, less than or equal to 0.1, or 0.01 to 0.3 at a frequency of 1 to 300 megahertz in the form of a solid ceramic.

(37) Aspect 9: The ferrite composition of any one or more of the preceding aspects, wherein the ferrite composition has a resonant frequency of greater than or equal to 250, or 250 to 500 megahertz in the form of a solid ceramic; or wherein the ferrite composition has a Snoek product of greater than or equal to 8 gigahertz, or greater than or equal to 9 gigahertz, or 9 to 15 gigahertz at a in the form of a solid ceramic.

(38) Aspect 10: A composite comprising a ferrite composition, for example, of any one or more of the preceding aspects and a polymer. The composite can be free of a void space.

(39) Aspect 11: An article comprising a ferrite composition, for example, of any one or more of Aspects 1 to 9 or the composite of Aspect 10.

(40) Aspect 12: The article of Aspect 11, wherein the article is an antenna, filter, inductor, circulator, or phase shifter.

(41) Aspect 13: The article of any one or more of Aspects 11 to 12, wherein the article is a microwave antenna.

(42) Aspect 14: The article of any one or more of Aspects 11 to 13, wherein the article is an antenna operable at a frequency of 1 to 300 megahertz, or 30 to 300 megahertz.

(43) Aspect 15: A method of making a ferrite composition, for example, of any one or more of the preceding aspects comprising: mixing ferrite precursor compounds comprising Me, Fe, Sb, Co, and M'; and sintering the ferrite precursor compounds in an oxygen atmosphere to form the SbCo—M-type ferrite.

(44) Aspect 16: The method of Aspect 15, wherein the ferrite precursor compounds comprise MeCO.sub.3 , Sb.sub.2O.sub.3 , Co.sub.3O.sub.4 , M'O.sub.2 , and Fe.sub.2O.sub.3 .

(45) Aspect 17: The method of Aspect 16, wherein the ferrite precursor compounds comprise 5 to 20 weight percent of MeCO.sub.3 , 0.01 to 20 weight percent of Sb.sub.2O.sub.3 , 1 to 15 weight percent of Co.sub.3O.sub.4 , 0.1 to 10 weight percent of M'O.sub.2 , and 50 to 80 weight percent of Fe.sub.2O.sub.3 ; all based on the total weight of the precursor compounds.

(46) Aspect 18: The method of any one or more of Aspects 15 to 17, wherein the sintering occurs at a sintering temperature of 1,000 to 1,300° C., or 1,200 to 1,250° C. for a sintering time of 1 to 20 hours, or 5 to 12 hours.

(47) Aspect 19: The method of any one or more of Aspects 15 to 18, further comprising calcining the ferrite precursor compounds in air prior to the sintering.

(48) Aspect 20: The method of any one or more of Aspects 15 to 19, further comprising mixing the ferrite precursor compounds with a binder prior to the sintering.

(49) Aspect 21: The method of any one or more of Aspects 15 to 20, further comprising forming the ferrite precursor compounds prior to the sintering.

(50) Aspect 22: A method of making a composite (for example, of Aspect 10) comprising mixing a ferrite composition (for example, of any one or more of Aspects 1 to 9 and optionally formed by the method of any one or more of Aspects 15 to 20) and a polymer to form the composite; and optionally shaping mixture; wherein the mixing optionally occurs at a temperature greater than or equal to the a melting temperature of the polymer.

(51) Aspect 23: A method of making a composite (for example, of Aspect 10) comprising mixing a ferrite composition (for example, of any one or more of Aspects 1 to 9 and optionally formed by the method of any one or more of Aspects 15 to 20) and a polymer precursor; and polymerizing the polymer precursor to form the composite.

(52) The compositions, methods, and articles can alternatively comprise, consist of, or consist essentially of, any appropriate materials, steps, or components herein disclosed. The compositions, methods, and articles can additionally, or alternatively, be formulated so as to be devoid, or

substantially free, of any materials (or species), steps, or components, that are otherwise not necessary to the achievement of the function or objectives of the compositions, methods, and articles.

(53) As used herein, “a,” “an,” “the,” and “at least one” do not denote a limitation of quantity, and are intended to cover both the singular and plural, unless the context clearly indicates otherwise. For example, “an element” has the same meaning as “at least one element,” unless the context clearly indicates otherwise. The term “at least one of” means that the list is inclusive of each element individually, as well as combinations of two or more elements of the list, and combinations of at least one element of the list with like elements not named. Also, the term “combination” is inclusive of blends, mixtures, alloys, reaction products, and the like. The term “or” means “and/or” unless clearly indicated otherwise by context. Reference throughout the specification to “an aspect”, “an embodiment”, “another embodiment”, “some embodiments”, and so forth, means that a particular element (e.g., feature, structure, step, or characteristic) described in connection with the embodiment is included in at least one embodiment described herein, and may or may not be present in other embodiments. In addition, it is to be understood that the described elements may be combined in any suitable manner in the various embodiments.

(54) Unless specified to the contrary herein, all test standards are the most recent standard in effect as of the filing date of this application, or, if priority is claimed, the filing date of the earliest priority application in which the test standard appears.

(55) The endpoints of all ranges directed to the same component or property are inclusive of the endpoints, are independently combinable, and include all intermediate points and ranges. For example, ranges of “up to 25 wt %, or 5 to 20 wt %” is inclusive of the endpoints and all intermediate values of the ranges of “5 to 25 wt %,” such as 10 to 23 wt %, etc.

(56) Unless defined otherwise, technical and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art to which this disclosure belongs.

(57) All cited patents, patent applications, and other references are incorporated herein by reference in their entirety. However, if a term in the present application contradicts or conflicts with a term in the incorporated reference, the term from the present application takes precedence over the conflicting term from the incorporated reference.

(58) While particular embodiments have been described, alternatives, modifications, variations, improvements, and substantial equivalents that are or may be presently unforeseen may arise to applicants or others skilled in the art. Accordingly, the appended claims as filed and as they may be amended are intended to embrace all such alternatives, modifications variations, improvements, and substantial equivalents.

Claims

1. A ferrite composition comprising a SbCo—M-type ferrite having the formula: $\text{Me}_{1-x}\text{Sb}_x\text{Co}_{y+x}\text{M}'_y\text{Fe}_{12-2x-2y}\text{O}_{19}$ wherein Me is at least one of Sr, Pb, or Ba; M' is at least one of Ti, Zr, Ru, or Ir; x is 0.001 to 0.3; and y is 1 to 1.2.
2. The ferrite composition of claim 1, wherein Me is at least Ba and/or wherein M' is at least Ti.
3. The ferrite composition of claim 1, wherein x is 0.005 to 0.1.
4. The ferrite composition of claim 1, wherein the ferrite composition has a permeability of greater than or equal to 30 at a frequency of 1 to 300 megahertz in the form of a solid ceramic.
5. The ferrite composition of claim 1, wherein the ferrite composition has a complex permeability at 200 megahertz of 30 to 40; or the SbCo—M-type ferrite can have a complex permeability at 300 megahertz of 20 to 35.
6. The ferrite composition of claim 1, wherein the ferrite composition has a magnetic loss tangent $\tan \delta_{\mu}$ of less than or equal to 0.3 at a frequency of 1 to 300 megahertz in the form of a solid ceramic.

7. The ferrite composition of claim 1, wherein the ferrite composition has a resonant frequency of greater than or equal to 250 megahertz in the form of a solid ceramic; or wherein the ferrite composition has a Snoek product of greater than or equal to 8 gigahertz in the form of a solid ceramic.

8. A composite comprising the ferrite composition of claim 1 and a polymer.

9. An article comprising the ferrite composition of claim 1.

10. An article comprising a ferrite composition comprising a SbCo—M-type ferrite having the formula:

$\text{Me}_{1-x}\text{Sb}_x\text{Co}_y\text{M}'_{1-y}\text{Fe}_{12-x-2y}\text{O}_{19}$ wherein Me is at least one of Sr, Pb, or Ba; M' is at least one of Ti, Zr, Ru, or In; x is 0.001 to 0.3; and y is 0.8 to 1.3, and wherein the article is an antenna, filter, inductor, circulator, or phase shifter.

11. The article of claim 9 wherein the article is a microwave antenna.

12. The article of claim 9, wherein the article is an antenna operable at a frequency of 1 to 300 megahertz.

13. A method of making a ferrite composition comprising a SbCo—M-type ferrite having the formula:

$\text{Me}_{1-x}\text{Sb}_x\text{Co}_y\text{M}'_{1-y}\text{Fe}_{12-x-2y}\text{O}_{19}$ wherein Me is at least one of Sr, Pb, or Ba; M' is at least one of Ti, Zr, Ru, or Ir; x is 0.001 to 0.3; and y is 0.8 to 1.3, the method comprising: mixing ferrite precursor compounds comprising Me, Fe, Sb, Co, and M'; and sintering the ferrite precursor compounds in an oxygen atmosphere to form the SbCo—M-type ferrite.

14. The method of claim 13, wherein the ferrite precursor compounds comprise MeCO_3 , Sb_2O_3 , Co_3O_4 , $\text{M}'\text{O}$, and Fe_2O_3 .

15. The method of claim 14, wherein the ferrite precursor compounds comprise 5 to 20 weight percent of MeCO_3 , 0.01 to 20 weight percent of Sb_2O_3 , 1 to 15 weight percent of Co_3O_4 , 0.1 to 10 weight percent of $\text{M}'\text{O}$, and 50 to 80 weight percent of Fe_2O_3 ; all based on the total weight of the precursor compounds.

16. The method of claim 13, wherein the sintering occurs at a sintering temperature of 1,000 to 1,300° C. for a sintering time of 1 to 20 hours.

17. The method of claim 13, further comprising calcining the ferrite precursor compounds in air prior to the sintering.

18. The method of claim 13, further comprising mixing the ferrite precursor compounds with a binder prior to the sintering.

19. The method of claim 13, further comprising forming the ferrite precursor compounds prior to the sintering.
